Conscription and Workplace Adjustment: Reallocation, Hiring, and Long-Run Career Effects

- Conflicts and wars displaces workers from their workplaces.
  Japanese bill in 1945 mobilized 65% of the working population into the battlefields.
- Replacement costs for employers soar during wartimes
  - Internal markets: Younger staff are often abscent due to conscription.
  - External markets: Shortage of labor supply.

Research Questions

- How are the vacancies filled by reassinging incumbent workers into different roles through different contracts?
- What are the long run effect of such temporarly reassignment on the career trajectories of the colleagues surrounding the draftee?

Identifying who filled draft-induced vacancies

- We study how draft-induced vacancies in a public-sector organization with more than 40,000 employees (e.g., the Tokyo civil service) were filled.
- We compare vacancy-filling patterns across organizationally adjacent sections in response to draft-induced employee exits.
  - The drafting process was undisclosed to the public $\Rightarrow$ drafting is as good as random after conditioning on occupation and section assignment.
- We compare how vacancies are filled through internal transfers versus external hiring.

Natural departures and replacement margins: peace time

Natural departures, peace time, 1930–1936

	Internal staff response				*External*
	Total internal	— origin of worker — Retention (same section)	Same dept. (cross-section)	Diff. dept. (cross-bureau)	New hires
Internal-transfer departures	-0.061	0.248***	-0.092**	-0.217***	-0.101***
	(0.080)	(0.076)	(0.042)	(0.054)	(0.035)
Exit departures	0.359*	0.812***	-0.108**	-0.344***	0.039
	(0.187)	(0.246)	(0.051)	(0.105)	(0.045)
Observations			2,871		
Internal-transfer departures			1,460		
Exit departures			1,237		
Year FE			Yes		
Section FE			Yes		
Position FE			Yes		
Log cumulative male baseline			Yes		

Notes: OLS regressions. Unit of observation: destination office × kyoku × ka × kakari × occupation × year. Each row is estimated separately and reports the coefficient on the listed number of male natural departures from the same full cell between $t-1$ and t. Internal-transfer departures are workers observed in a different cell in t; exit departures are workers absent from the personnel directory in t. Draft-coded workers are excluded. Outcomes use the same replacement-margin definitions as the draft table. Standard errors clustered by office in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

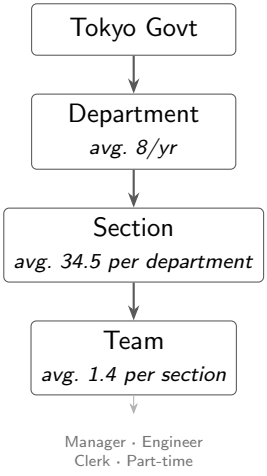
Natural departures and replacement margins: war time

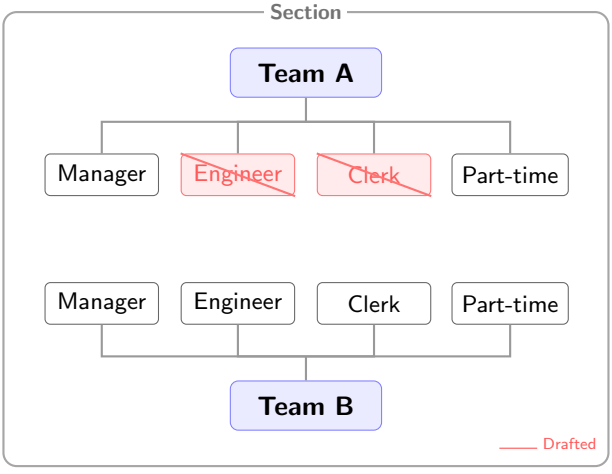
Natural departures, war time, 1937–1943

	Internal staff response				*External*
	Total internal	— origin of worker — Retention (same section)	Same dept. (cross-section)	Diff. dept. (cross-bureau)	New hires
Internal-transfer departures	-0.141	-0.201**	0.099***	-0.039**	-0.097*
	(0.101)	(0.081)	(0.038)	(0.018)	(0.055)
Exit departures	0.154	0.135	0.068	-0.050***	-0.047
	(0.151)	(0.167)	(0.069)	(0.015)	(0.071)
Observations			7,470		
Internal-transfer departures			2,667		
Exit departures			3,203		
Year FE			Yes		
Section FE			Yes		
Position FE			Yes		
Log cumulative male baseline			Yes		

Notes: OLS regressions. Unit of observation: destination office × kyoku × ka × kakari × occupation × year. Each row is estimated separately and reports the coefficient on the listed number of male natural departures from the same full cell between $t-1$ and t. Internal-transfer departures are workers observed in a different cell in t; exit departures are workers absent from the personnel directory in t. Draft-coded workers are excluded. Outcomes use the same replacement-margin definitions as the draft table. Standard errors clustered by office in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

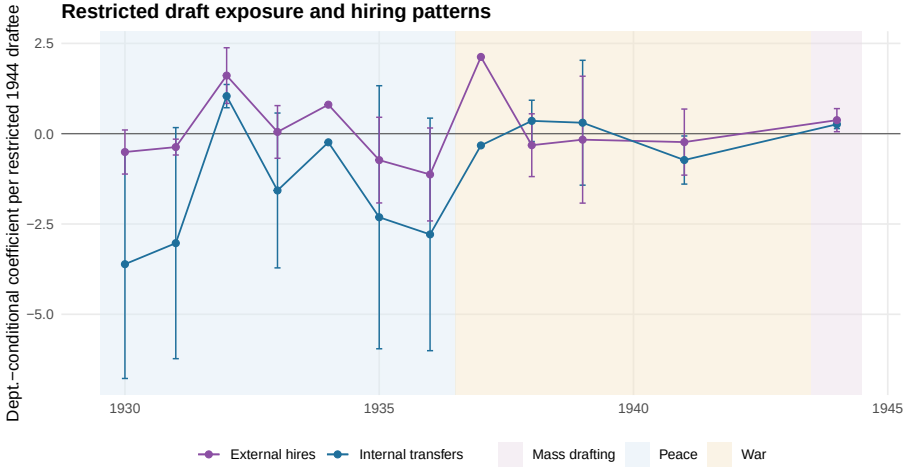
Identification: within-section, within-occupation





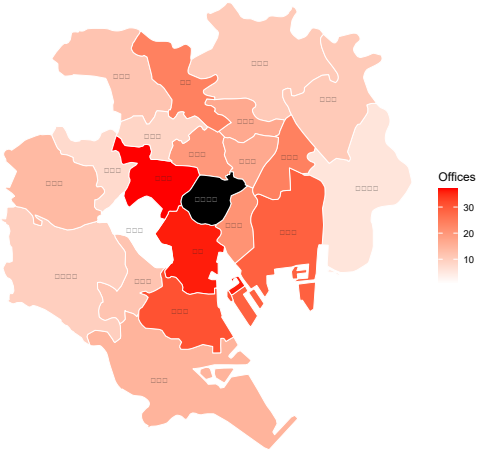
Balance test (backup)

Restricted exposure: baseline yearly correlations



Unit: office × team × occupation cells in offices with any restricted 1944 draft exposure. Estimates for 1930–1943 are baseline correlations; 1944 is the mass-drafting response year. Treatment is the 1944 male post-merger entrant draft count at the office × occupation level. Regressions condition on department and occupation and control for log cumulative male baseline.

Tokyo offices across wards



Empirical strategy: OLS with fixed effects

- For each hiring/transfer margin, we estimate:

$$Y_{okt} = \beta \cdot \text{Draftees}_{okt} + \theta \cdot \text{Workers}_{kt} + \alpha_o + \gamma_k + \varepsilon_{okt}$$

- Y_{okt}: hiring or transfer margin in occupation o, section k, and year t.
- Draftees_{okt}: number of drafted workers.
- Workers_{kt}: number of workers in section k in year t.
- α_o: occupation fixed effects.
- γ_k: section fixed effects.

Drafting and replacement margins

Composition of the 2:1 ratio (internal vs. external)

	Internal staff response ≈ **0.41**				*External* ≈ **0.21**
	Total internal	— origin of worker — Retention (same section)	Same dept. (cross-section)	Diff. dept. (cross-bureau)	New hires
No. drafted workers	0.4079***	0.2269***	0.1001**	0.0809***	0.2057**
	(0.0761)	(0.0730)	(0.0403)	(0.0154)	(0.0998)
Ratio: internal / external = 0.41 / 0.21 ≈ **2 : 1**					
Observations	7,409				
Year FE	Yes				
Section FE	Yes				
Position FE	Yes				
Log cumulative male baseline	Yes				

Notes: OLS regressions. Unit of observation: destination office × kakari × occupation × year, 1938–1945. Each cell reports the coefficient on the number of drafted male workers in the same destination kakari-occupation-year. "Retention" counts workers who remained in the same ka group (same-section stayers). "Same dept." counts workers from a different ka group within the same normalized kyoku group (cross-section transfers). "Diff. dept." counts workers from a different normalized kyoku group (cross-bureau transfers). "Total internal" is the sum of all three. New hires exclude first observed office-years. Standard errors clustered by office in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Matching appendix table

Post-1943 entrant drafting and replacement margins

Composition using post-1943 entrant draftees

	Internal staff response ≈ **0.47**				*External* ≈ **0.34**
	Total internal	— origin of worker — Retention (same section)	Same dept. (cross-section)	Diff. dept. (cross-bureau)	New hires
No. drafted post-1943 entrants	0.465*	0.221	0.096	0.148***	0.335
	(0.246)	(0.188)	(0.070)	(0.042)	(0.243)
Ratio: internal / external = 0.47 / 0.34 ≈ **1.4 : 1**					
Observations			7,409		
Entrant-drafted workers in supported cells			389		
Year FE			Yes		
Section FE			Yes		
Position FE			Yes		
Log cumulative male baseline			Yes		

Notes: OLS regressions. Unit of observation: destination office × kakari × occupation × year, 1938–1945. Each cell reports the coefficient on the number of male post-1943 entrants drafted in the same destination kakari-occupation-year. "Retention" counts workers who remained in the same ka group (same-section stayers). "Same dept." counts workers from a different ka group within the same normalized kyoku group (cross-section transfers). "Diff. dept." counts workers from a different normalized kyoku group (cross-bureau transfers). "Total internal" is the sum of all three. New hires exclude first observed office-years. Standard errors clustered by office in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Appendix

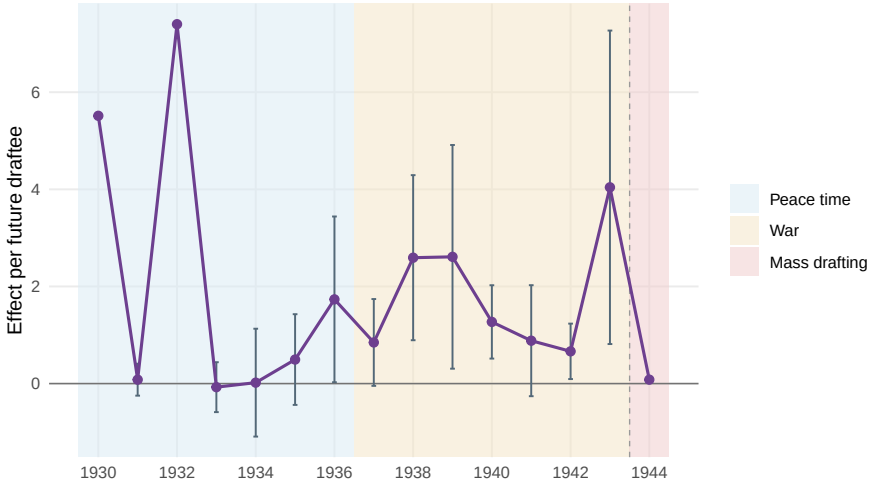
Draft intensity: internal transfers DID

	Section + occ. + year FE	No section FE	No occ. FE
No. future drafted × War	2.098***	2.320***	2.060***
	(0.678)	(0.812)	(0.654)
No. future drafted × Mass drafting	-0.120	0.004	0.001
	(0.152)	(0.133)	(0.058)
Observations	936	988	938
Sections clustered		173	
Baseline male employment (log)	Yes	Yes	Yes

Notes: Outcome is total internal arrivals in the section × team × occupation cell. Treatment is the number of male workers drafted in 1944 from the same future cell. Peace time, 1930–1936, is the omitted period; War is 1937–1943; Mass drafting is 1944. Standard errors clustered by section in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Appendix table | Appendix ES plot | Full decomposition

Draft intensity: internal transfers event study



Outcome: total internal arrivals. Treatment: number of male workers drafted in 1944 from the same future cell. Conditional on section and occupation; controls for log cumulative male baseline. Bars show 95% confidence intervals.

Appendix table | Appendix ES plot | Full decomposition

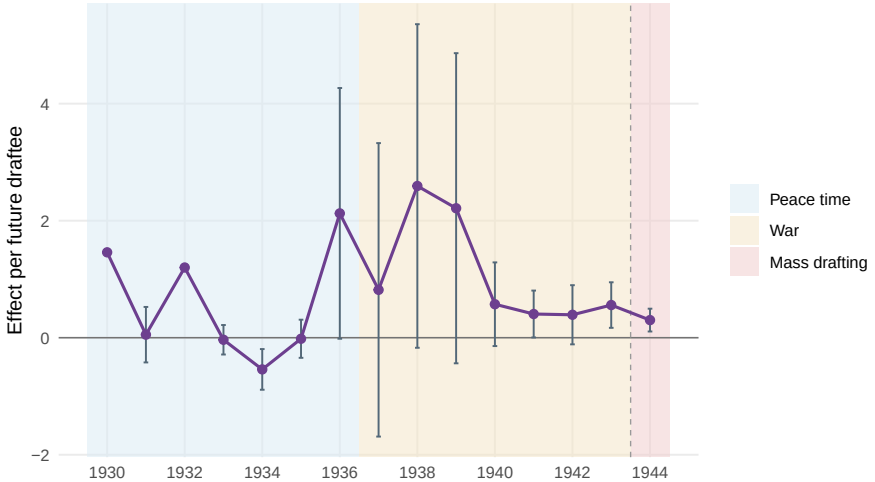
Draft intensity: external hires DID

	Section + occ. + year FE	No section FE	No occ. FE
No. future drafted × War	0.924	0.903*	1.014*
	(0.562)	(0.523)	(0.580)
No. future drafted × Mass drafting	0.261**	0.302***	0.310***
	(0.107)	(0.089)	(0.103)
Observations	936	988	938
Sections clustered		173	
Baseline male employment (log)	Yes	Yes	Yes

Notes: Outcome is external new hires in the section × team × occupation cell. Treatment is the number of male workers drafted in 1944 from the same future cell. Peace time, 1930–1936, is the omitted period; War is 1937–1943; Mass drafting is 1944. Standard errors clustered by section in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Appendix table | Appendix ES plot | Full decomposition

Draft intensity: external hires event study



Outcome: external new hires. Treatment: number of male workers drafted in 1944 from the same future cell. Conditional on section and occupation; controls for log cumulative male baseline. Bars show 95% confidence intervals.

Appendix table | Appendix ES plot | Full decomposition

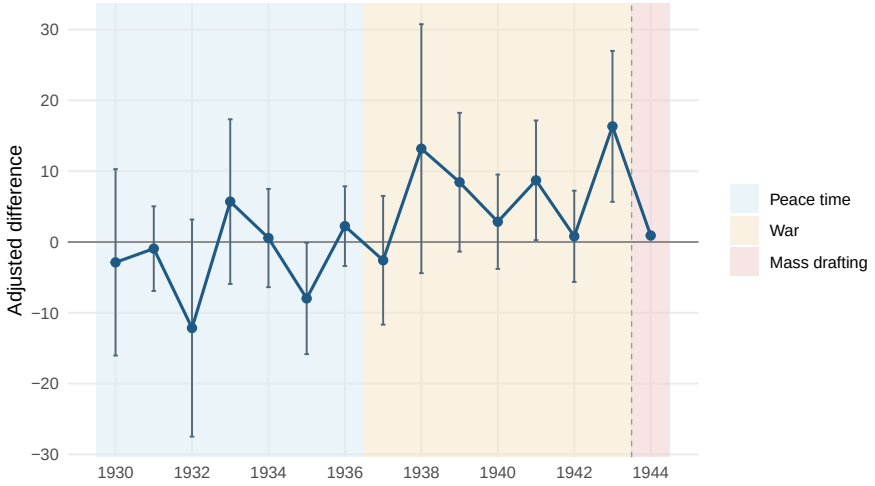
Internal transfers: period DID

	Section + occ. + year FE	No section FE	No occ. FE
Future drafted cell × War	9.880***	7.596***	13.275***
	(2.964)	(2.920)	(3.011)
Future drafted cell × Mass drafting	0.117	1.240***	1.652***
	(0.411)	(0.353)	(0.409)
Observations	10,108	10,286	10,111
Sections clustered		982	
Baseline male employment (log)	Yes	Yes	Yes

Notes: Outcome is total internal arrivals in the section × team × occupation cell, excluding workers later marked drafted in the 1944 TokyoTo directory. Treatment is whether the cell contains any 1944 male draftee. Peace time, 1930–1936, is the omitted period; War is 1937–1943; Mass drafting is 1944. Standard errors clustered by section in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Same-dept table | Same-dept ES plot | Full decomposition

Internal transfers: event study



Unit: section × team × occupation. Conditional on section and occupation; controls for log cumulative male baseline. Bars show 95% confidence intervals.

Same-dept table | Same-dept ES plot | Full decomposition

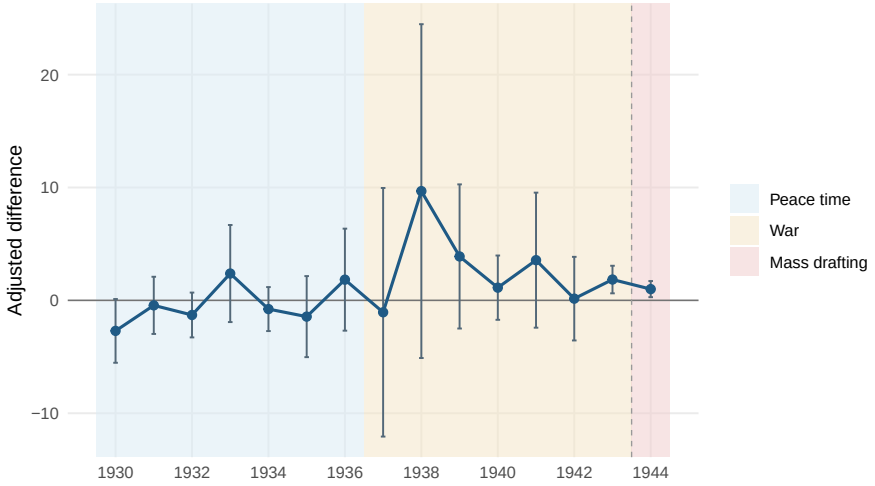
External hires: period DID

	Section + occ. + year FE	No section FE	No occ. FE
Future drafted cell × War	2.861	2.322	4.499**
	(1.741)	(1.645)	(1.764)
Future drafted cell × Mass drafting	-0.223	0.740**	0.949***
	(0.384)	(0.361)	(0.358)
Observations	10,108	10,286	10,111
Sections clustered		982	
Baseline male employment (log)	Yes	Yes	Yes

Notes: Outcome is external new hires in the section × team × occupation cell, excluding workers later marked drafted in the 1944 TokyoTo directory. Treatment is whether the cell contains any 1944 male draftee. Peace time, 1930–1936, is the omitted period; War is 1937–1943; Mass drafting is 1944. Standard errors clustered by section in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

New-hire table | New-hire ES plot | Full decomposition

External hires: event study



Unit: section × team × occupation. Conditional on section and occupation; controls for log cumulative male baseline. Bars show 95% confidence intervals.

New-hire table | New-hire ES plot | Full decomposition

Peace-time replacement timing

Tokyo-Shi baseline, 1929–1937

	Internal staff response				*External*
	Total internal	— decomposition — Retention (same section)	Same dept. (cross-section)	Diff. dept. (cross-bureau)	New hires
Response in $t-1$	0.5934***	0.2206***	0.2362**	0.1367***	0.2191***
	(0.0723)	(0.0616)	(0.1085)	(0.0426)	(0.0448)
Response in t	0.0258	0.1774**	-0.0583*	-0.0933**	0.0054
	(0.0692)	(0.0789)	(0.0312)	(0.0364)	(0.0244)
Response in $t+1$	-0.0642	-0.0161	-0.0561*	0.0081	-0.0160
	(0.0643)	(0.0436)	(0.0340)	(0.0100)	(0.0138)
Observations			3,950		
Natural male cell departures			3,742		
Year FE			Yes		
Section FE			Yes		
Position FE			Yes		
Baseline male employment (log)			Yes		

Notes: Each row regresses replacement margins in the indicated year on natural male cell departures measured between $t-1$ and t in the same full office-kyoku-ka-kakari-occupation cell. The positive $t-1$ row indicates that peace-time departures are preceded by larger routine inflows; the contemporaneous and $t+1$ rows show little evidence of backfilling after the departure. Standard errors clustered by office in parentheses. $^{***}p<0.01$, $^{**}p<0.05$, $^{*}p<0.1$.

Bilateral mobility: nested organizational distance

Unit: origin office $\times$ origin occupation $\times$ destination office $\times$ destination occupation $\times$ year.

$$E[F_{op,dq,t}] = \exp\{\beta_1 \log N_{op,t-1} + \beta_2 \log N_{dq,t-1} + \beta_3 \log N_{op,t-1} \log N_{dq,t-1} + \beta_4 \text{OccSimilarity}_{od,t-1} + \beta_5 \mathbf{1}\{\text{same kyoku, diff. ka}\} + \beta_6 \mathbf{1}\{\text{diff. kyoku}\} + \alpha_t +$$

- Retention is same office and same occupation; excluded from the flow panel.
- Baseline category: workers change occupation within the same ka $\times$ kyoku.
- **Same kyoku, different ka**: within-department cross-section mobility.
- **Different kyoku**: cross-department mobility. Sample excludes office-years with drafting at either endpoint.

Bilateral mobility: strong nested distance gradient

	(1) Year FE	(2) Occ-pair FE	(3) Origin kyoku FE	(4) Dest. kyoku FE	(5) Origin ka FE	(6) Dest. ka FE
log $N_{o,p}$	0.778***	0.861***	0.773***	0.781***	0.855***	0.763***
	(0.075)	(0.099)	(0.077)	(0.076)	(0.074)	(0.092)
log $N_{d,q}$	0.239***	0.130	0.218***	0.222***	0.163**	0.151**
	(0.060)	(0.099)	(0.062)	(0.064)	(0.072)	(0.066)
log $N_{o,p} \times$ log $N_{d,q}$	0.007	-0.041	0.012	0.010	0.036	0.031
	(0.019)	(0.027)	(0.019)	(0.019)	(0.025)	(0.025)
Occupation similarity	1.788***	1.090***	1.536***	1.404***	1.574***	1.296***
	(0.359)	(0.381)	(0.366)	(0.384)	(0.371)	(0.422)
Same kyoku, diff. ka	-2.334***	-3.519***	-2.324***	-2.375***	-2.361***	-2.408***
	(0.326)	(0.339)	(0.346)	(0.358)	(0.340)	(0.344)
Different kyoku	-4.659***	-5.889***	-4.860***	-4.943***	-4.897***	-5.025***
	(0.271)	(0.272)	(0.298)	(0.287)	(0.290)	(0.290)
Observations	4,445,908	2,684,319	4,406,699	3,590,943	2,778,413	1,420,716
Pseudo R^2	0.333	0.416	0.352	0.357	0.389	0.410

Notes: PPML regressions. Unit is origin office–occupation $\times$ destination office–occupation $\times$ year, 1938–1945. Retention, defined as same office and same occupation, is excluded. Omitted category is same office (ka $\times$ kyoku), different occupation. A literal same-kakari baseline has no positive flow cells in the kakari-covered sample. Standard errors clustered two-way by origin and destination office. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Selection into draft-induced vacancies

	All transfers		Different ka, same kyoku		Different kyoku	
	Any: Draft Vac.	Any: Non-Draft	Diff Ka: Draft	Diff Ka: Non-Draft	Diff Kyoku: Draft	Diff Kyoku: Non-Draft
Decoration rank (0–8; high=better)	-0.0529	0.2466	-15.4652***	0.2863	0.1670	0.2020
	(0.4393)	(0.1728)	(0.0000)	(0.1787)	(0.4425)	(0.2030)
Court rank (0–8; high=better)	-0.0186	-0.1181**	-0.3238	-0.0864	0.0372	-0.1504**
	(0.1355)	(0.0601)	(0.3940)	(0.0667)	(0.1420)	(0.0734)
Female	-1.0508	0.3935*	-18.5681***	0.4252*	-0.6950	0.3173
	(0.8136)	(0.2358)	(0.0000)	(0.2467)	(0.8277)	(0.2850)
Tenure (yrs)	0.1014***	0.0295**	-0.0950**	-0.0209	0.2123***	0.0965***
	(0.0223)	(0.0125)	(0.0414)	(0.0139)	(0.0258)	(0.0145)
Log salary (prior year)	-0.1404	-0.2396***	0.1765	-0.1763*	-0.3365*	-0.3171***
	(0.1625)	(0.0921)	(0.2320)	(0.1012)	(0.1891)	(0.1050)
Year FE			Yes			
Observations			5,523			
Log-likelihood	-3140.5				-5295.7	
AIC	6329.1				10687.5	

Sample restricted to origin office-unit × year cells with at least one transfer to a different ka or kyoku, further restricted to workers with observed prior-year salary. Base category: stayed in same ka and kyoku (required for MNL interpretation). Rank variables are coded 0–8, with higher values indicating higher status.

Trajectory regression: transferees vs peers

	Tenure after	Postwar surv.	Postwar years	Rank gain
Transferred vs. workers who **stayed at donor**	3.7525***	0.1953***	0.6345***	0.1813***
	(0.0911)	(0.0052)	(0.0206)	(0.0071)
Draft-induced vs. **natural-exit** destination	1.5513***	0.0801***	0.4697***	0.0304**
	(0.2350)	(0.0153)	(0.0649)	(0.0137)
Transferred vs. workers **already at destination**	2.2992***	0.1426***	0.4895***	0.1348***
	(0.0868)	(0.0059)	(0.0208)	(0.0060)
Fixed-effects				
Team	Yes	Yes	Yes	Yes
Occupation	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes
Observations	Row 1: 91,586	Row 2: 40,060	Row 3: 97,145	

*Clustered SE in parentheses; ***: 0.01, **: 0.05, *: 0.1*

Promotion and bonus salary responses

	No. promoted workers		*No. bonus recipients*	
Model:	(1)	(2)	(3)	(4)
Variables				
No. drafted workers	0.0267*	0.0231*	0.1367**	0.1548***
	(0.0107)	(0.0114)	(0.0529)	(0.0587)
No. drafted × Part-time		0.0154		−0.1854**
		(0.0112)		(0.0610)
Fixed-effects				
Year / Section / Occupation	Yes	Yes		
Section / Occupation			Yes	Yes
Log section size	Yes	Yes	Yes	Yes
Observations	2,770	2,770	576	576

*Clustered SE in parentheses; ***: 0.01, **: 0.05, *: 0.1*

Further analysis

- How were promotions and bonus pay used to transfer or retain incumbent workers?
  Findings: Both were used more in impacted offices. Promotions were used more often together with transfers.
- How did early promotion help transferred workers in their subsequent careers?
  Findings:
  1. Transferees sent to drafted offices outperform those sent to non-drafted offices.
  2. Transferees from remote teams perform worse than transferees from neighboring teams.

Discussion and conclusion

- Workplaces compensated for staff losses jointly through internal and external markets (about 2:1 per drafted employee).
- These results support empirical work on responses to worker exits (Jaeger, Heining, and Lazarus, 2025; the worker-death paper).
  But we show that replacement costs varied with the composition of workers surrounding each affected worker.
- Transferred workers presumably accelerated promotion and contributed to better career trajectories.

References I

Empirical strategy: who is selected for transfer?

For each donor office unit-year, worker i chooses among transfer outcomes $j \in \{\text{stay}, \text{draft vacancy}, \text{non-draft vacancy}\}$:

$$\Pr(y_{it} = j \mid X_{it}) = \frac{\exp\left(X_{it}'\beta_j\right)}{\sum_{\ell \in \mathcal{J}} \exp\left(X_{it}'\beta_\ell\right)}, \qquad X_{it}'\beta_{\text{stay}} = 0.$$

$$X_{it}'\beta_j = \beta_{0j} + \beta_{1j}\text{Decor}_{i,t-1} + \beta_{2j}\text{Court}_{i,t-1} + \beta_{3j}\text{Female}_{it} + \beta_{4j}\text{Tenure}_{it} + \beta_{5j}\log(\text{Salary}_{i,t-1} + 1) + \lambda_{jt}.$$

- Outcomes are estimated separately for leaving at all, within-kyoku transfers, and cross-kyoku transfers.
- Decoration and CourtRank are coded 0–8, with 0 for unobserved status.
- Female is an indicator, Tenure is years since first observed, and λ_{jt} are outcome-specific year fixed effects.

Drafting and replacement margins: geographic benchmark

Panel A. Composition of the 2:1 ratio (internal vs. external)

	Internal staff response ≈ **0.41**				*External* ≈ **0.21**
	Total internal	— decomposition — Retention (same section)	Same dept. (cross-section)	Diff. dept. (cross-bureau)	New hires
No. drafted workers	0.4079***	0.2269***	0.1001**	0.0809***	0.2057**
	(0.0761)	(0.0730)	(0.0403)	(0.0154)	(0.0998)

Ratio: internal / external = 0.41 / 0.21 ≈ **2 : 1**

Panel B. Geographic benchmark for internal replacements

	Total internal	≤2 km	>2 km	Unknown dist.	New hires
No. drafted workers	0.4079***	0.2506***	0.0780	0.0793	0.2057**
	(0.0761)	(0.0860)	(0.0529)	(0.0642)	(0.0998)
Observations			7,409		
Year FE			Yes		
Section FE			Yes		
Position FE			Yes		
Log cumulative male baseline			Yes		

Notes: OLS regressions. Unit of observation: destination office × kakari × occupation × year, 1938–1945. Each cell reports the coefficient on the number of drafted male workers in the same destination kakari-occupation-year. "Retention" counts workers who remained in the same ka group (same-section stayers). "Same dept." counts workers from a different ka group within the same normalized kyoku group (cross-section transfers). "Diff. dept." counts workers from a different normalized kyoku group (cross-bureau transfers). "Total internal" is the sum of all three. Panel B splits internal replacements by great-circle distance between prior-year and current-year office ward centroids. New hires exclude first observed office-years. All specifications include year, section, and position fixed effects and control for log cumulative male baseline employment. Standard errors clustered by office in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Peace-time cell departures and replacement margins

Tokyo-Shi baseline, 1929–1937

	Internal staff response				*External*
	Total internal	— decomposition — Retention (same section)	Same dept. (cross-section)	Diff. dept. (cross-bureau)	New hires
Natural male cell departures	0.0258	0.1774**	-0.0583*	-0.0933**	0.0054
	(0.0692)	(0.0789)	(0.0312)	(0.0364)	(0.0244)
Observations			3,950		
Natural male cell departures			3,742		
Year FE			Yes		
Section FE			Yes		
Position FE			Yes		
Baseline male employment (log)			Yes		

Notes: OLS regressions using Tokyo-Shi directory transitions, 1929–1937. Unit of observation: destination office × kyoku × ka × kakari × occupation × year. Observations are regression cell-years, not transfer counts. The treatment is the number of male workers in the same full cell in $t-1$ who are no longer in that exact cell in year t, excluding draft-coded rows. Outcomes use the same replacement-margin definitions as the wartime draft table: retention within the same ka group, transfers from a different ka group within the same normalized kyoku group, transfers from a different normalized kyoku group, and new hires excluding first observed office-years. All specifications include year, section, and position fixed effects and control for log cumulative male baseline employment. Standard errors clustered by office in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

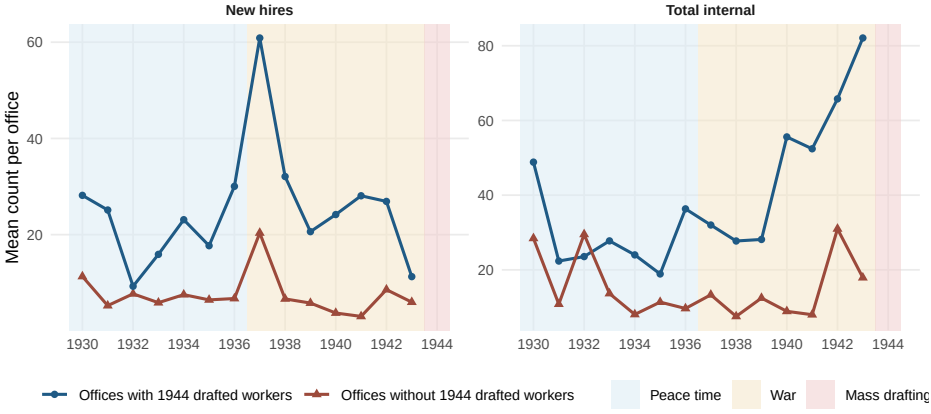
Placebo: future drafting and prior replacements

Pre-trend test: 1943 outcomes, 1944 draftees

	Internal staff response				*External*
		— decomposition —			
	Total internal	Retention (same section)	Same dept. (cross-section)	Diff. dept. (cross-bureau)	New hires
Future drafted workers in 1944	2.7463***	1.6695***	1.0425	0.0344**	0.3301***
	(0.7759)	(0.4830)	(0.6364)	(0.0157)	(0.1216)
Observations			803		
Future drafted workers			821		
Section FE			Yes		
Position FE			Yes		
Baseline male employment (log)			Yes		

Notes: OLS regressions. Outcomes are replacement margins in 1943, measured before the 1944 draft shock and excluding workers who are drafted in 1944 from the 1943 outcome counts. Treatment is the number of male workers drafted in 1944, assigned to the worker's actual 1943 office-kyoku-ka-kakari-occupation cell. The test asks whether future draft intensity predicts prior replacement patterns. Specifications include section and position fixed effects and control for log cumulative male baseline employment. Standard errors clustered by office in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Office pre-trajectories: drafted offices



Descriptive

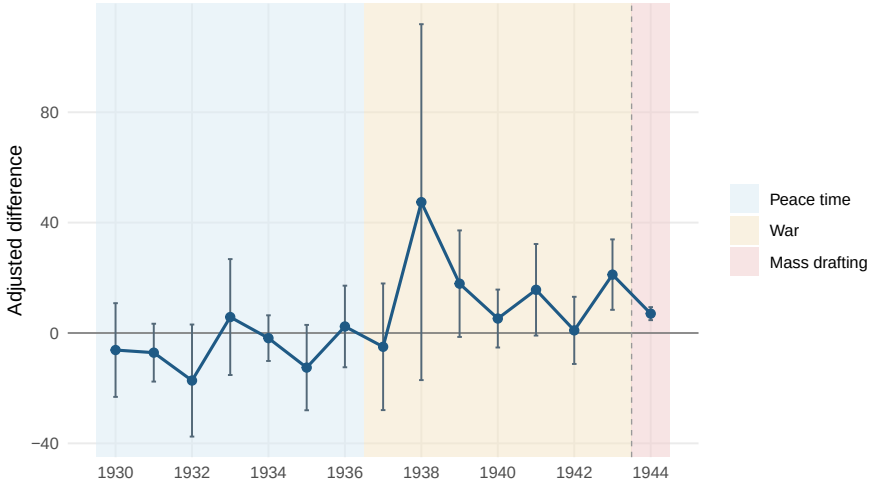
office-level 1930–1943 trajectories, grouped by whether the office contains any worker later marked drafted in the 1944 TokyoTo directory. Outcomes exclude those eventual draftees.

Pre-trends: adjusted office differences

	Adjusted difference: offices with 1944 drafted workers					
	1930	1933	1936	1939	1941	1943
Total internal	16.929	12.029	12.760*	11.945**	42.652***	58.037***
	(19.476)	(8.022)	(7.512)	(5.707)	(11.499)	(10.456)
Retention	-4.544*	0.450	9.348	2.730	22.992***	31.079***
	(2.353)	(0.478)	(6.298)	(1.889)	(7.217)	(8.077)
New hires	13.382*	8.536*	12.197*	9.128**	21.595***	3.225**
	(8.047)	(4.913)	(7.005)	(4.503)	(6.647)	(1.636)
Future-drafted offices				118		
Future drafted workers				554		
Baseline male employment (log)				Yes		

Notes: Separate year-by-year office-level OLS differences relative to offices without future draftees. Outcomes exclude workers later marked drafted in the 1944 TokyoTo directory. The table reports selected years from the 1930–1943 trajectory window. Specifications control for log cumulative male baseline employment. Standard errors clustered by office in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Within-section pre-trend: cell size



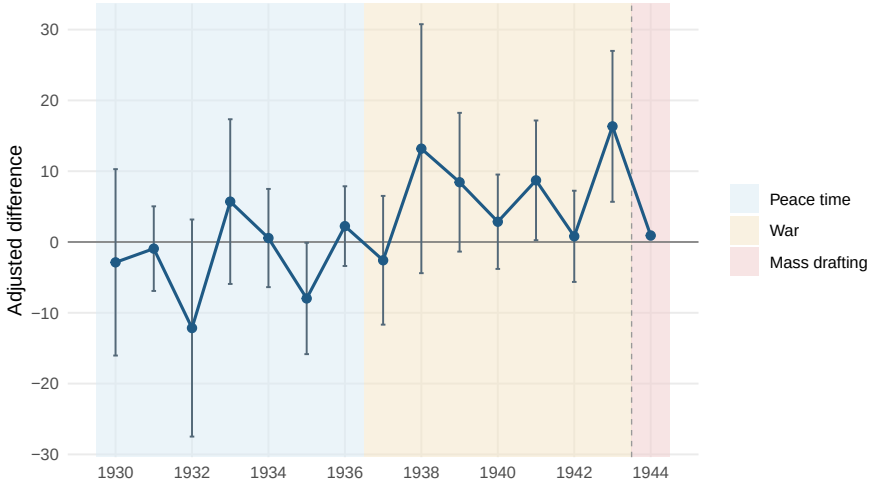
Unit: section × team × occupation. Conditional on section and position; controls for log cumulative male baseline. Bars show 95% confidence intervals.

Within-section pre-trend: cell size DID

	Cell + year FE	Section/position + year FE
Future drafted cell × War	29.468***	16.411***
	(7.660)	(5.901)
Future drafted cell × Mass drafting	1.568	4.114***
	(9.395)	(1.303)
Observations	6,776	10,108
Cells	5,419	
Sections	982	
Baseline male employment (log)	Yes	Yes

Notes: Outcome is measured in the section × team × occupation cell, excluding workers later marked drafted in the 1944 TokyoTo directory. Treatment is whether the cell contains any 1944 male draftee. Coefficients are DID estimates relative to peace time. Peace time, 1930–1936, is the omitted period; War is 1937–1943; Mass drafting is 1944. Standard errors clustered by section in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Within-section pre-trend: internal flows



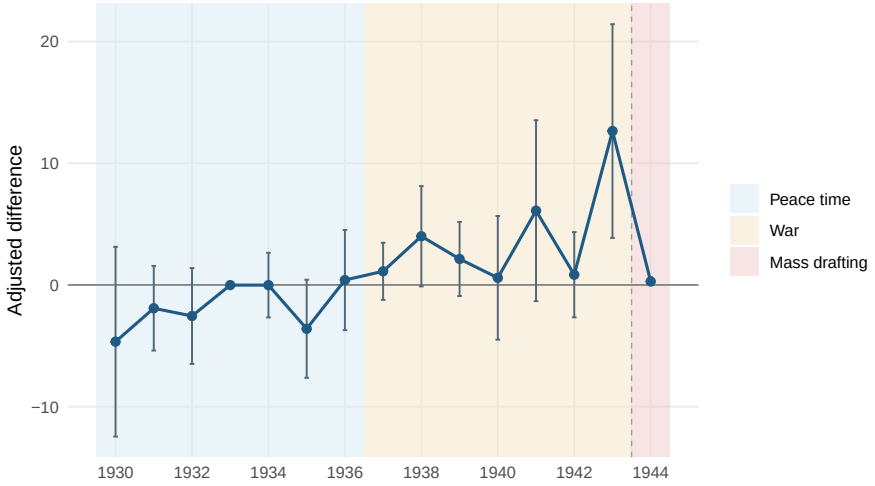
Unit: section × team × occupation. Conditional on section and position; controls for log cumulative male baseline. Bars show 95% confidence intervals.

Within-section pre-trend: internal flows DID

	Cell + year FE	Section/position + year FE
Future drafted cell × War	12.967***	9.880***
	(3.938)	(2.964)
Future drafted cell × Mass drafting	-1.042	0.117
	(4.310)	(0.411)
Observations	6,776	10,108
Cells		5,419
Sections		982
Baseline male employment (log)	Yes	Yes

Notes: Outcome is measured in the section × team × occupation cell, excluding workers later marked drafted in the 1944 TokyoTo directory. Treatment is whether the cell contains any 1944 male draftee. Coefficients are DID estimates relative to peace time. Peace time, 1930–1936, is the omitted period; War is 1937–1943; Mass drafting is 1944. Standard errors clustered by section in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Within-section pre-trend: retention



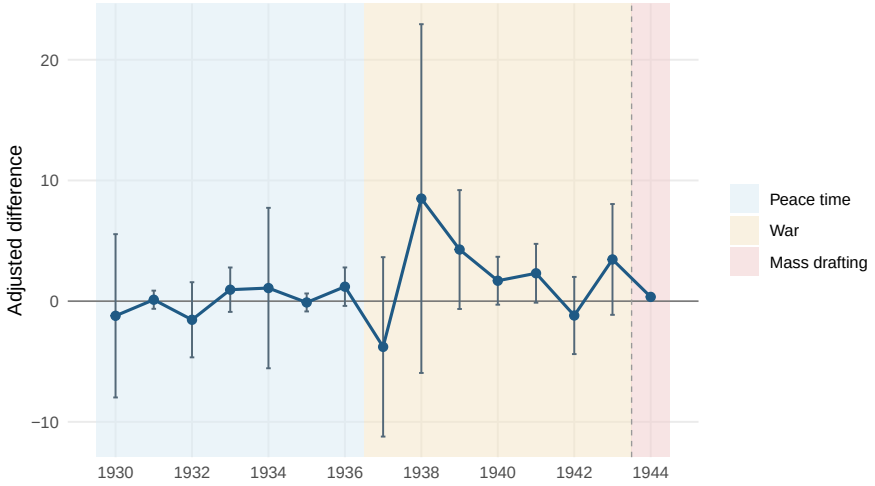
Unit: section × team × occupation. Conditional on section and position; controls for log cumulative male baseline. Bars show 95% confidence intervals.

Within-section pre-trend: retention DID

	Cell + year FE	Section/position + year FE
Future drafted cell × War	8.373***	6.754***
	(2.520)	(1.835)
Future drafted cell × Mass drafting	2.219**	0.219
	(1.053)	(0.213)
Observations	6,776	10,108
Cells	5,419	
Sections	982	
Baseline male employment (log)	Yes	Yes

Notes: Outcome is measured in the section × team × occupation cell, excluding workers later marked drafted in the 1944 TokyoTo directory. Treatment is whether the cell contains any 1944 male draftee. Coefficients are DID estimates relative to peace time. Peace time, 1930–1936, is the omitted period; War is 1937–1943; Mass drafting is 1944. Standard errors clustered by section in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Within-section pre-trend: same-dept transfers



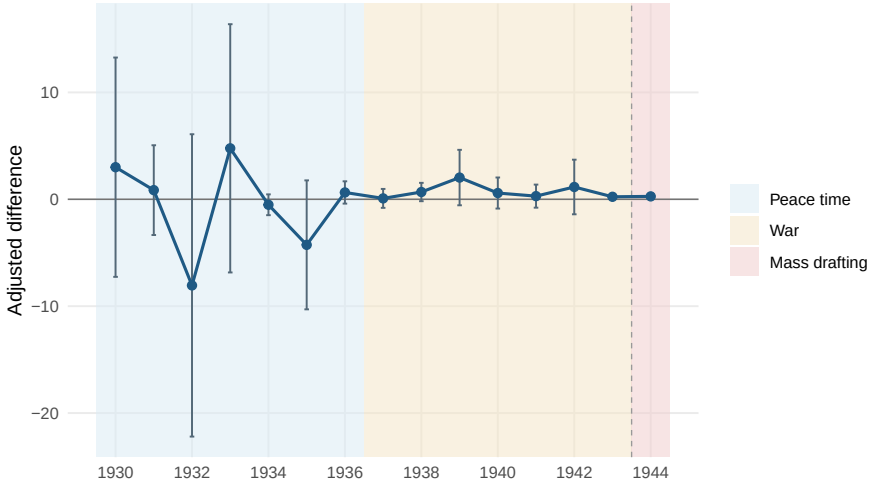
Unit: section × team × occupation. Conditional on section and position; controls for log cumulative male baseline. Bars show 95% confidence intervals.

Within-section pre-trend: same-dept transfers DID

	Cell + year FE	Section/position + year FE
Future drafted cell × War	2.849	2.759*
	(1.731)	(1.416)
Future drafted cell × Mass drafting	-0.722	-0.237
	(1.231)	(0.215)
Observations	6,776	10,108
Cells		5,419
Sections		982
Baseline male employment (log)	Yes	Yes

Notes: Outcome is measured in the section × team × occupation cell, excluding workers later marked drafted in the 1944 TokyoTo directory. Treatment is whether the cell contains any 1944 male draftee. Coefficients are DID estimates relative to peace time. Peace time, 1930–1936, is the omitted period; War is 1937–1943; Mass drafting is 1944. Standard errors clustered by section in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Within-section pre-trend: diff.-dept transfers



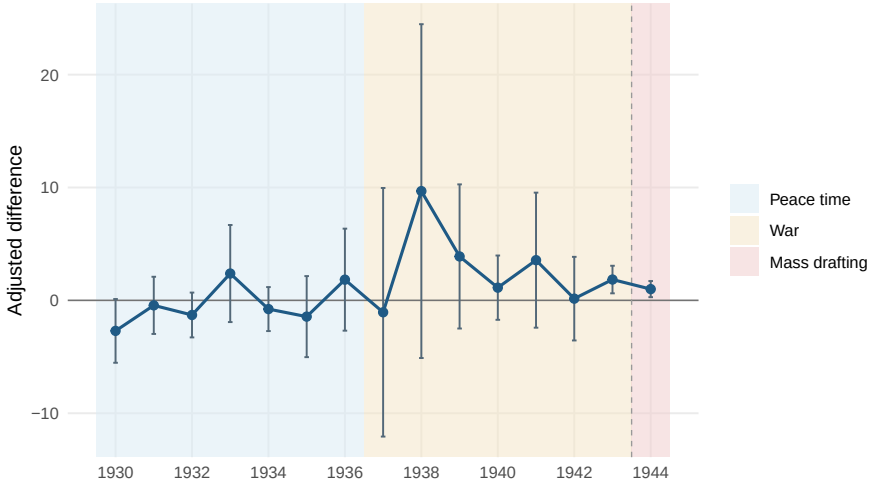
Unit: section × team × occupation. Conditional on section and position; controls for log cumulative male baseline. Bars show 95% confidence intervals.

Within-section pre-trend: diff.-dept transfers DID

	Cell + year FE	Section/position + year FE
Future drafted cell × War	1.744***	0.366
	(0.648)	(0.421)
Future drafted cell × Mass drafting	-2.539	0.135
	(3.512)	(0.176)
Observations	6,776	10,108
Cells	5,419	
Sections	982	
Baseline male employment (log)	Yes	Yes

Notes: Outcome is measured in the section × team × occupation cell, excluding workers later marked drafted in the 1944 TokyoTo directory. Treatment is whether the cell contains any 1944 male draftee. Coefficients are DID estimates relative to peace time. Peace time, 1930–1936, is the omitted period; War is 1937–1943; Mass drafting is 1944. Standard errors clustered by section in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Within-section pre-trend: new hires



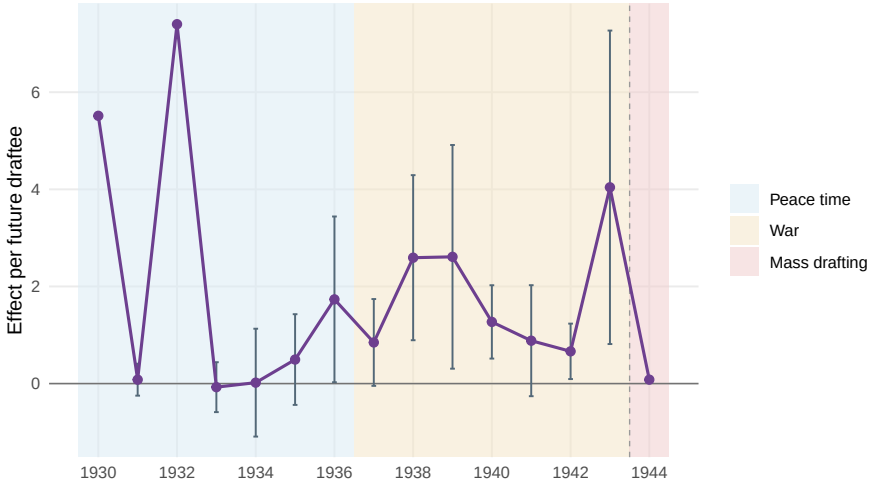
Unit: section × team × occupation. Conditional on section and position; controls for log cumulative male baseline. Bars show 95% confidence intervals.

Within-section pre-trend: new hires DID

	Cell + year FE	Section/position + year FE
Future drafted cell × War	7.137**	2.861
	(2.908)	(1.741)
Future drafted cell × Mass drafting	2.037	-0.223
	(2.395)	(0.384)
Observations	6,776	10,108
Cells	5,419	
Sections	982	
Baseline male employment (log)	Yes	Yes

Notes: Outcome is measured in the section × team × occupation cell, excluding workers later marked drafted in the 1944 TokyoTo directory. Treatment is whether the cell contains any 1944 male draftee. Coefficients are DID estimates relative to peace time. Peace time, 1930–1936, is the omitted period; War is 1937–1943; Mass drafting is 1944. Standard errors clustered by section in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Draft intensity: internal transfers event study



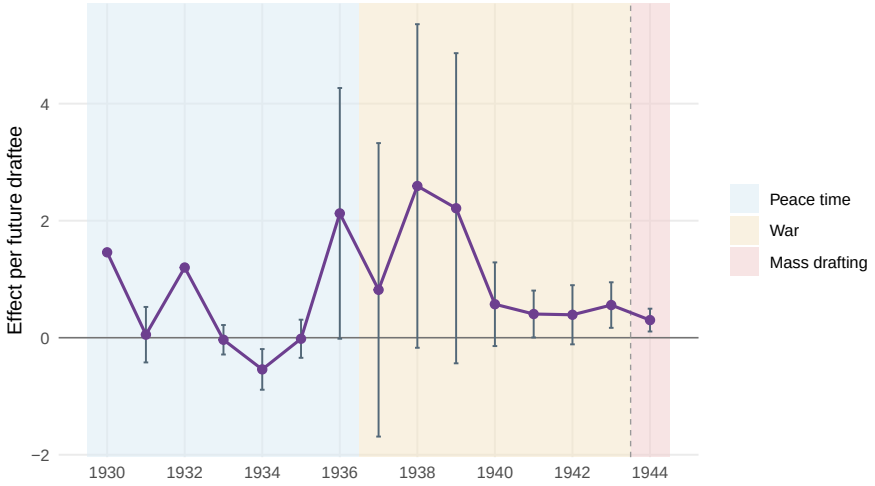
Outcome: total internal arrivals; treatment: number of future draftees. Conditional on section and occupation; controls for log cumulative male baseline. Bars show 95% confidence intervals.

Draft intensity: internal transfers event study DID

	Section + occ. + year FE	No section FE	No occ. FE
No. future drafted × War	2.098***	2.320***	2.060***
	(0.678)	(0.812)	(0.654)
No. future drafted × Mass drafting	-0.120	0.004	0.001
	(0.152)	(0.133)	(0.058)
Observations	936	988	938
Sections clustered		173	
Baseline male employment (log)	Yes	Yes	Yes

Notes: Outcome is measured in the section × team × occupation cell. Treatment is the number of male workers drafted in 1944 from the same future cell. Peace time, 1930–1936, is the omitted period; War is 1937–1943; Mass drafting is 1944. Standard errors clustered by section in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Draft intensity: external hires event study



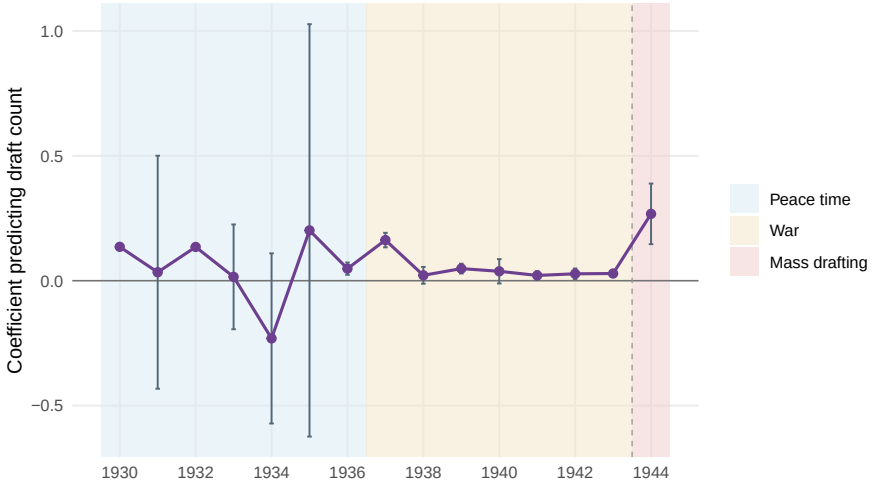
Outcome: external new hires; treatment: number of future draftees. Conditional on section and occupation; controls for log cumulative male baseline. Bars show 95% confidence intervals.

Draft intensity: external hires event study DID

	Section + occ. + year FE	No section FE	No occ. FE
No. future drafted × War	0.924	0.903*	1.014*
	(0.562)	(0.523)	(0.580)
No. future drafted × Mass drafting	0.261**	0.302***	0.310***
	(0.107)	(0.089)	(0.103)
Observations	936	988	938
Sections clustered		173	
Baseline male employment (log)	Yes	Yes	Yes

Notes: Outcome is measured in the section × team × occupation cell. Treatment is the number of male workers drafted in 1944 from the same future cell. Peace time, 1930–1936, is the omitted period; War is 1937–1943; Mass drafting is 1944. Standard errors clustered by section in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Draft intensity orthogonality: cell size



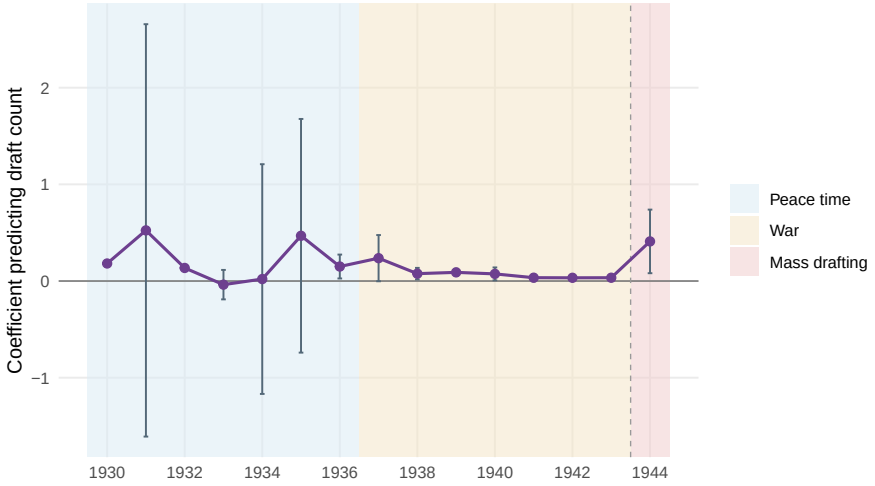
Predictor: workers in team × occupation cell. Sample: units with at least one 1944 draftee. Dependent variable is 1944 draft count. Conditional on section and position; controls for log cumulative male baseline. Bars show 95% confidence intervals.

Draft intensity orthogonality: cell size DID

	Cell + year FE	Section/position + year FE
Workers × War	-0.000	-0.003
	(0.000)	(0.008)
Workers × Mass drafting	-0.155***	0.264***
	(0.036)	(0.076)
Observations	652	936
Cells	475	
Sections	173	
Baseline male employment (log)	Yes	Yes

Notes: Dependent variable is 1944 draft count in units with at least one 1944 draftee. Coefficients report whether the listed margin predicts draft intensity differently in war or mass-drafting years relative to peace time. Peace time, 1930–1936, is the omitted period; War is 1937–1943; Mass drafting is 1944. Standard errors clustered by section in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Draft intensity orthogonality: internal flows



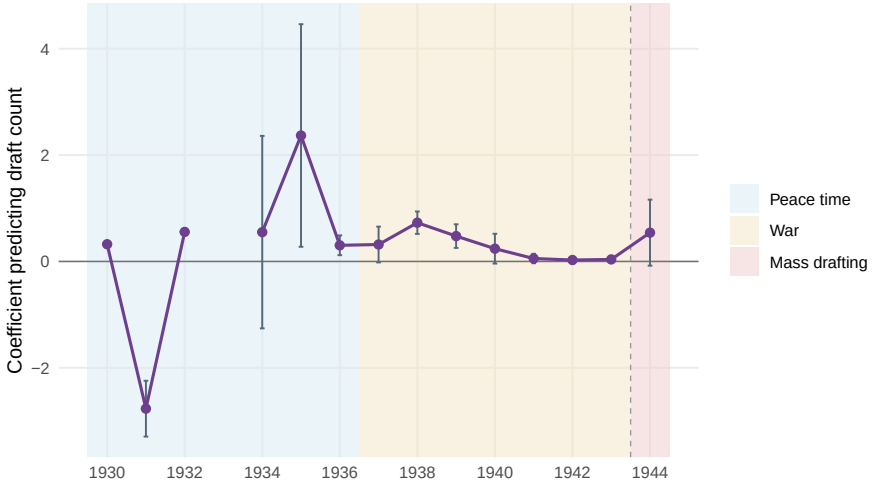
Predictor: total internal arrivals. Sample: units with at least one 1944 draftee. Dependent variable is 1944 draft count. Conditional on section and position; controls for log cumulative male baseline. Bars show 95% confidence intervals.

Draft intensity orthogonality: internal flows DID

	Cell + year FE	Section/position + year FE
Total internal × War	0.000	0.008
	(0.000)	(0.020)
Total internal × Mass drafting	-3.972***	0.313**
	(1.444)	(0.143)
Observations	652	936
Cells	475	
Sections	173	
Baseline male employment (log)	Yes	Yes

Notes: Dependent variable is 1944 draft count in units with at least one 1944 draftee. Coefficients report whether the listed margin predicts draft intensity differently in war or mass-drafting years relative to peace time. Peace time, 1930–1936, is the omitted period; War is 1937–1943; Mass drafting is 1944. Standard errors clustered by section in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Draft intensity orthogonality: retention



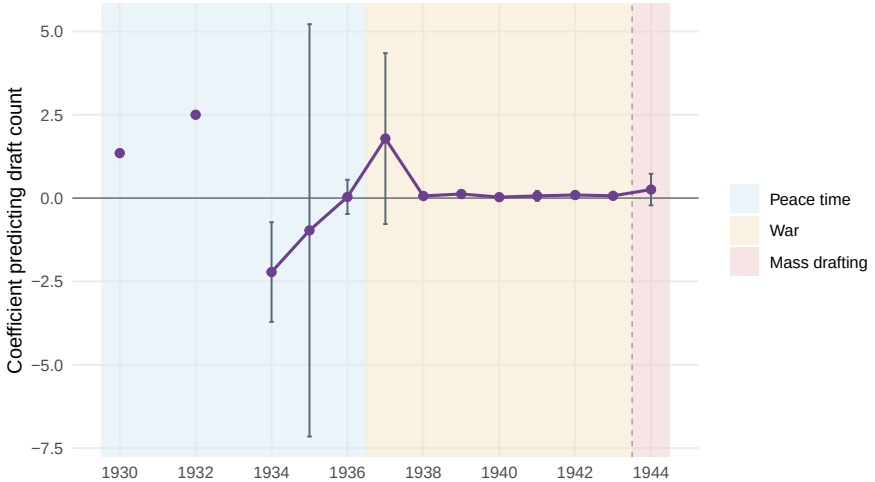
Predictor: same-section retention. Sample: units with at least one 1944 draftee. Dependent variable is 1944 draft count. Conditional on section and position; controls for log cumulative male baseline. Bars show 95% confidence intervals.

Draft intensity orthogonality: retention DID

	Cell + year FE	Section/position + year FE
Retention × War	-0.001**	-0.074
	(0.000)	(0.058)
Retention × Mass drafting	-3.971***	0.307
	(1.444)	(0.268)
Observations	652	936
Cells	475	
Sections	173	
Baseline male employment (log)	Yes	Yes

Notes: Dependent variable is 1944 draft count in units with at least one 1944 draftee. Coefficients report whether the listed margin predicts draft intensity differently in war or mass-drafting years relative to peace time. Peace time, 1930–1936, is the omitted period; War is 1937–1943; Mass drafting is 1944. Standard errors clustered by section in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Draft intensity orthogonality: same-dept transfers



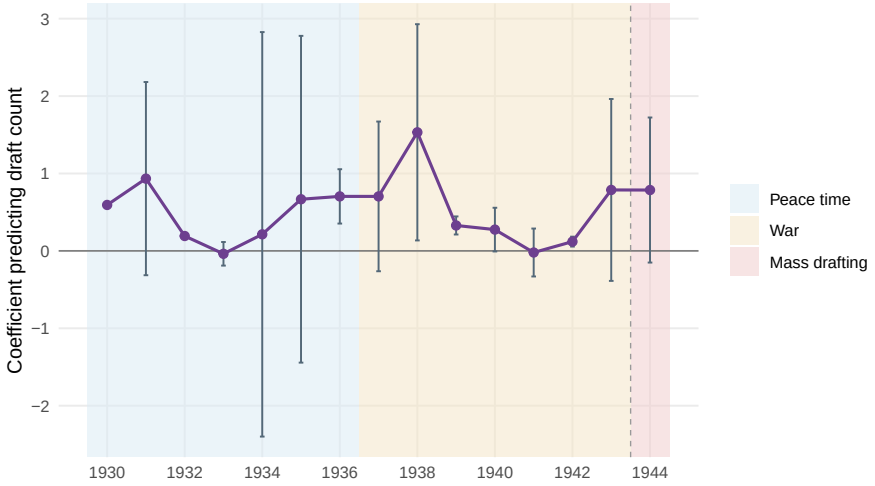
Predictor: cross-section transfers within department. Sample: units with at least one 1944 draftee. Dependent variable is 1944 draft count. Conditional on section and position; controls for log cumulative male baseline. Bars show 95% confidence intervals.

Draft intensity orthogonality: same-dept transfers DID

	Cell + year FE	Section/position + year FE
Same dept. × War	0.000	0.024
	(0.000)	(0.031)
Same dept. × Mass drafting	NA	0.236
	NA	(0.235)
Observations	652	936
Cells	475	
Sections	173	
Baseline male employment (log)	Yes	Yes

Notes: Dependent variable is 1944 draft count in units with at least one 1944 draftee. Coefficients report whether the listed margin predicts draft intensity differently in war or mass-drafting years relative to peace time. Peace time, 1930–1936, is the omitted period; War is 1937–1943; Mass drafting is 1944. Standard errors clustered by section in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Draft intensity orthogonality: diff.-dept transfers



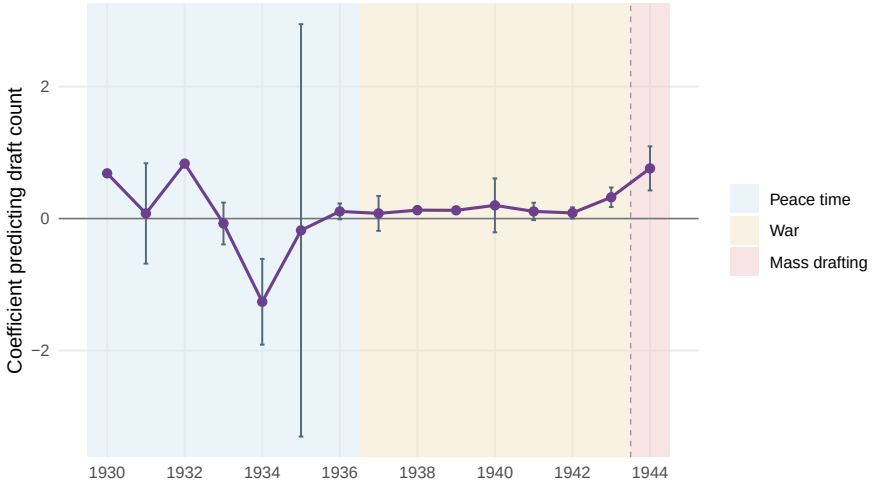
Predictor: cross-department transfers. Sample: units with at least one 1944 draftee. Dependent variable is 1944 draft count. Conditional on section and position; controls for log cumulative male baseline. Bars show 95% confidence intervals.

Draft intensity orthogonality: diff.-dept transfers DID

	Cell + year FE	Section/position + year FE
Diff. dept. × War	-0.000	0.110**
	(0.001)	(0.049)
Diff. dept. × Mass drafting	NA	0.664*
	NA	(0.397)
Observations	652	936
Cells	475	
Sections	173	
Baseline male employment (log)	Yes	Yes

Notes: Dependent variable is 1944 draft count in units with at least one 1944 draftee. Coefficients report whether the listed margin predicts draft intensity differently in war or mass-drafting years relative to peace time. Peace time, 1930–1936, is the omitted period; War is 1937–1943; Mass drafting is 1944. Standard errors clustered by section in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Draft intensity orthogonality: new hires



Predictor: external new hires. Sample: units with at least one 1944 draftee. Dependent variable is 1944 draft count. Conditional on section and position; controls for log cumulative male baseline. Bars show 95% confidence intervals.

Draft intensity orthogonality: new hires DID

	Cell + year FE	Section/position + year FE
New hires × War	0.001	-0.001
	(0.001)	(0.033)
New hires × Mass drafting	-0.777***	0.727***
	(0.252)	(0.212)
Observations	652	936
Cells	475	
Sections	173	
Baseline male employment (log)	Yes	Yes

Notes: Dependent variable is 1944 draft count in units with at least one 1944 draftee. Coefficients report whether the listed margin predicts draft intensity differently in war or mass-drafting years relative to peace time. Peace time, 1930–1936, is the omitted period; War is 1937–1943; Mass drafting is 1944. Standard errors clustered by section in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Restricted exposure: baseline period correlations

	Peace, 1930–1936	War, 1937–1943
Internal transfers	-2.236**	-0.450
	(0.940)	(1.347)
External hires	-0.393	0.031
	(0.393)	(0.445)
Retention	-1.450*	-0.443
	(0.764)	(0.946)
Same-dept transfers	-0.434	0.038
	(0.566)	(0.347)
Diff.-dept transfers	-0.352	-0.046
	(0.238)	(0.092)
Observations	353	398
Departments clustered	21	32
Baseline male employment (log)	Yes	Yes
Department, occupation, year FE	Yes	Yes

Notes: Unit: baseline office × team × occupation × year cells, restricted to offices with any male post-merger entrant drafted in 1944. Treatment is the 1944 restricted draft count at the office × occupation level. Coefficients report department-conditional baseline correlations per restricted 1944 draftee. Standard errors clustered by department. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Restricted exposure: support

Baseline office × team × occupation × year rows	762
Offices with restricted 1944 draft exposure	21
Departments observed in baseline panel	40
Baseline office × team × occupation cells	154
Exposed baseline office × team × occupation cells	11
Exposed 1944 office × occupation cells	11
Male post-merger entrant draftees in exposure measure	601

Notes: Support for the department-conditional baseline correlation exercise. Exposure is measured in 1944 among male post-merger entrants and projected to baseline office × team × occupation outcomes through office × occupation cells.

Balance test of treatment and control

Outcome (pre-war)	Extensive coef.	p-value	Intensive coef.	p-value
No. new hires	-2.05	0.094	-0.0987	0.446
	(1.15)		(0.1263)	
No. internal transfers	40.5	0.533	1.1205	0.025
	(63.5)		(0.4533)	
No. promoted	0.464	0.055	0.0132	0.146
	(0.224)		(0.0087)	
Female share	-0.0014	0.859	-0.0002	0.152
	(0.0080)		(0.0001)	
Sample size (obs.)	571 (outcome-specific in full table)			
Controls	FE: Section + position + year; + log team-cell size			

Back to identification slide

Research question idea: Tokyo bombing shock

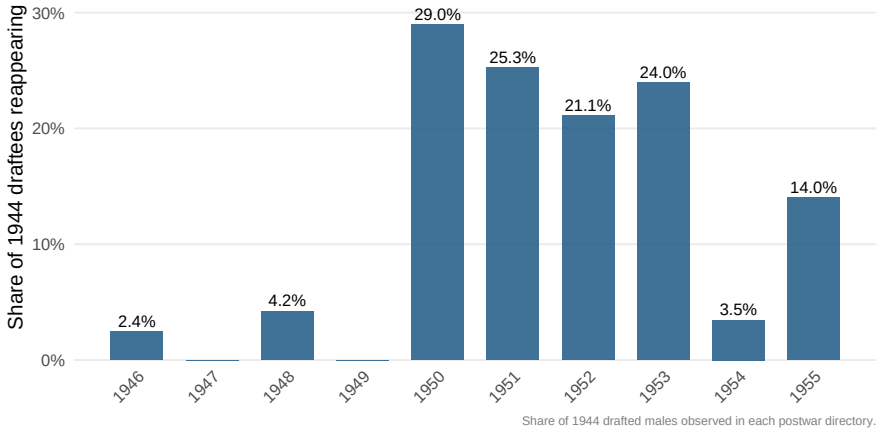
- How did the 1945 air-raid displace managers from teams?
- In 1947, Tokyo removed the bar for women to take managerial positions.
  $\Rightarrow$ Number of female managers increased from 10 to 100 in 1948.
- Did women managers get assigned to teams with a displaced manager?
- Did women managers retain/promote female employees more than male employees?

Did drafted workers return after the war?

	Basic OLS	Kyoku FE	Section FE
Drafted in 1944	0.4215***	0.4216***	0.4331***
	(0.0124)	(0.0158)	(0.0169)
Drafted return rate		42.3%	
Non-drafted exiter return rate		0.0%	
Drafted workers		1,849	
Non-drafted exiters (1944)		3,794	
Kyoku FE	No	Yes	—
Section FE	No	No	Yes
Controls (tenure, rank)	Yes	Yes	Yes

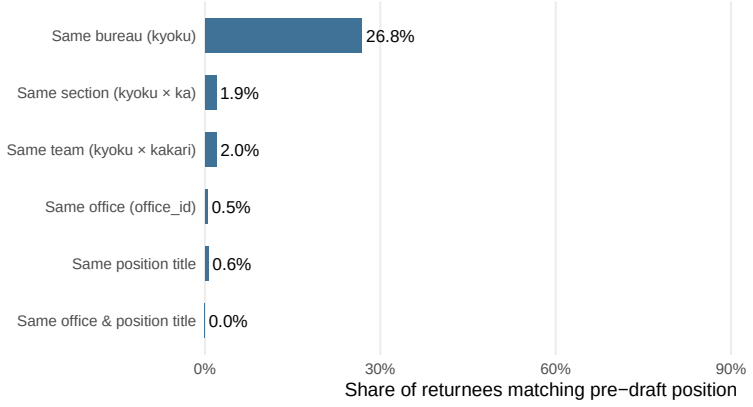
Notes: OLS regression. Outcome is an indicator equal to 1 if the worker is observed in any postwar directory (1947–1955). Sample is male workers whose last observed year is 1944 ("exiters"), so both groups left in 1944. "Drafted" workers are those flagged as drafted in the 1944 TokyoTo directory; the comparison group left for other reasons. Controls include years of tenure in 1944 and position rank. Standard errors clustered by office. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

When did drafted workers return?



Share of 1944 drafted males (N = 1,849) observed in each postwar Tokyo metropolitan government directory. No 1947 or 1949 directories available. The large spike in 1950 likely reflects the completion of demobilization and occupation-era restructuring.

Returnees: did they come back to the same position?



Among the 782 drafted workers who returned to Tokyo metropolitan government employment (1947–1955), the figure shows what share returned to the same organizational unit at each level of the hierarchy. Only 26.8% returned to the same bureau; virtually none (0.5%) returned to the same office. The 1943 Tokyo Fu/Shi merger and postwar reorganization largely dissolved the original structure.

Same-bureau return and postwar tenure

	Postwar years observed	
	(1) No FE	(2) Bureau FE
Same bureau (kyoku)	-0.018	0.227
	(0.111)	(0.153)
Bureau match rate	26.8% (N = 508 with non-missing kyoku)	
Observations	508	507
Bureau FE (pre-draft kyoku)	No	Yes
Pre-draft rank + years until return	Yes	Yes

Notes: OLS regressions. Sample: male 1944 draftees who returned to Tokyo metropolitan government employment in 1947–1955 (N = 782). "Same bureau" is an indicator equal to 1 if the worker's first postwar kyoku matches their 1944 kyoku. Match rates at finer levels (section, office, position title) are near zero (0.5–1.9%) due to the 1943 Fu/Shi merger and postwar reorganization. "Postwar yrs" counts postwar directories (1947–1955) in which the worker appears. Controls include years between 1944 and first postwar appearance and pre-draft position rank. Standard errors clustered by pre-draft office. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.